ADVANCED DATA MINING

Customer Booking Behavior Analysis using Classification and Clustering

British Airways Dataset - Forage

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CONTENTS

CHAPTER 1

BUSINESS PROCESS DESCRIPTION → TRANSACTION MAPPING

1.1 Business Process Description

The business process analysed in this case study is **customer booking behaviour prediction in the airline/travel domain**.

Customer booking completion is influenced by multiple factors such as travel preferences, booking channel, trip type, and timing. Understanding these patterns helps organizations improve customer targeting, optimize marketing strategies, and increase booking conversions.

This study applies:

- **Classification algorithms** to predict booking completion
- **Clustering techniques** to identify customer segments

1.2 Transaction and Item Identification

- **Transaction:** Each row represents a customer booking instance
- **Items (Features):**
  - Sales Channel
  - Trip Type
  - Flight Day
  - Route
  - Booking Origin
  - Other numerical attributes

1.3 Transaction Mapping

Raw Attribute Values	Transaction Representation
Online, RoundTrip, Monday	{Sales_Channel_Online, Trip_RoundTrip, Flight_Monday}

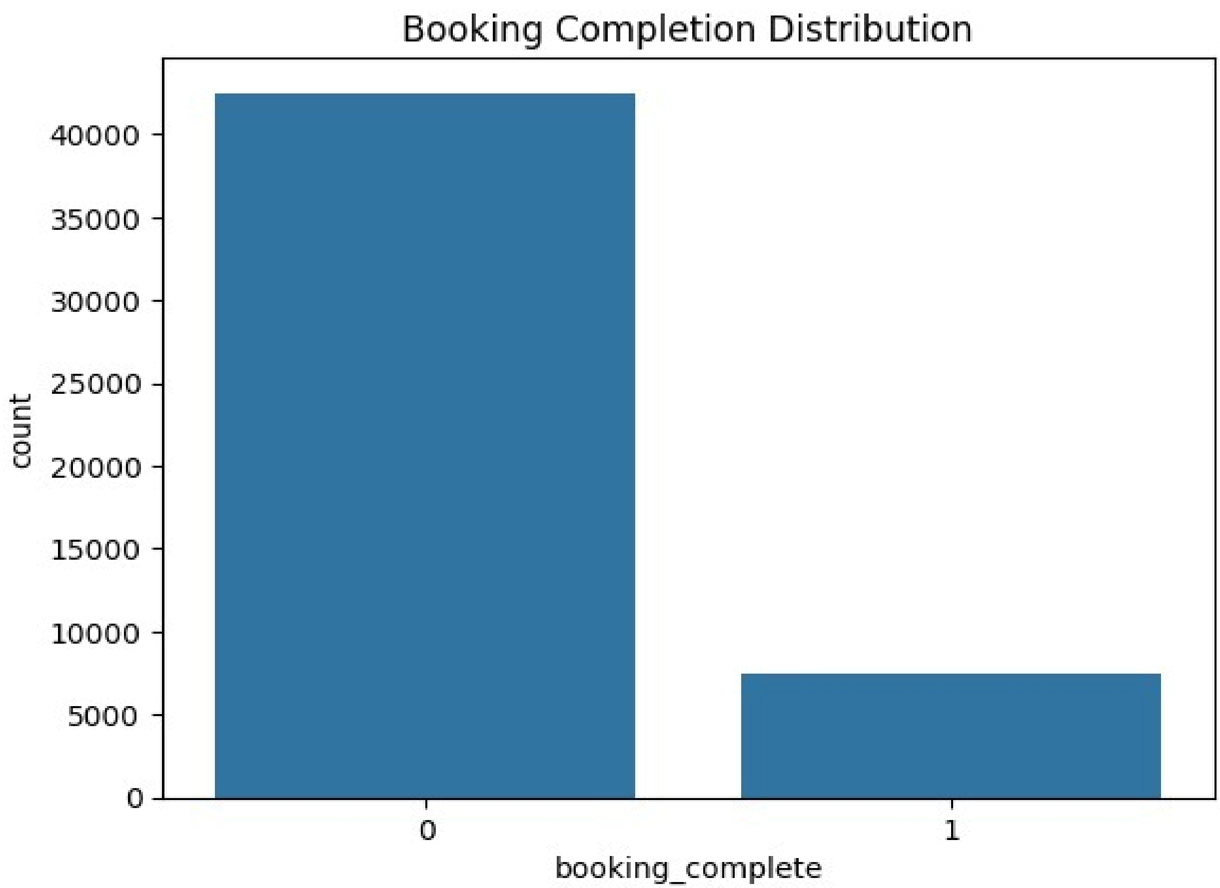
This transformation ensures compatibility with machine learning models.

CHAPTER 2
DATASET DESCRIPTION

- Dataset Type: Customer Booking Dataset (Forage)
- Number of Records**:** (50,000 rows)
- Number of Attributes: Multiple categorical and numerical features
- Target Variable: Booking Completion (0 / 1)

2.1 Data Understanding

The dataset contains customer booking and travel-related information used to predict whether a booking is completed.



CHAPTER 3

JUSTIFICATION OF MODEL PARAMETERS

3.1 Data Preprocessing

- Label Encoding applied to categorical features
- Missing values handled (if any)

Inference

Ensures structured input for machine learning models.

3.2 Feature Scaling

StandardScaler was used.

Why important?

- Required for **KNN & SVM**
- Prevents dominance of large-scale features

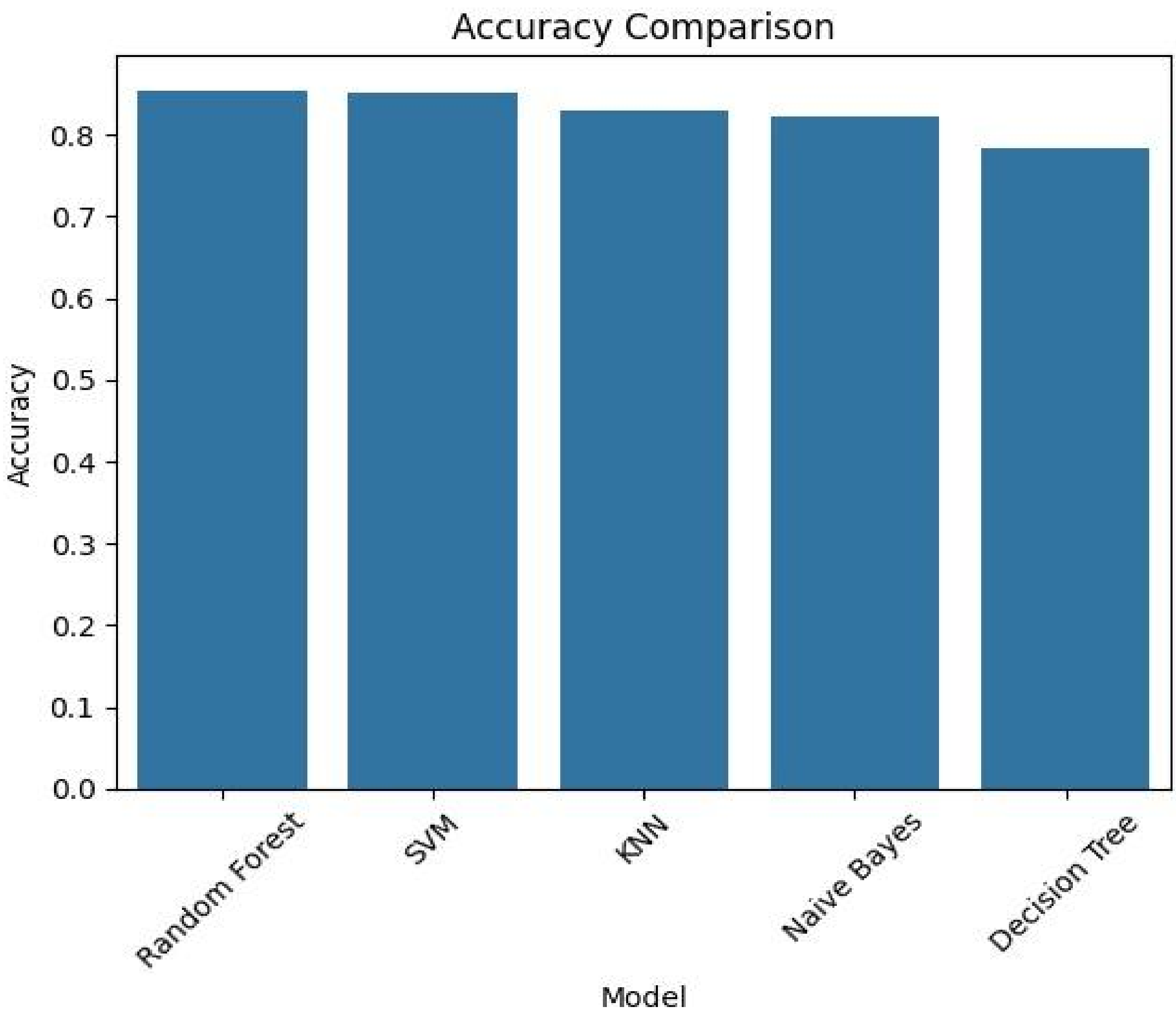
3.3 Model Selection Rationale

Model	Reason
Decision Tree	Interpretability
KNN	Distance-based learning
Naive Bayes	Fast probabilistic model
SVM	Handles complex boundaries
Random Forest	High accuracy + robustness

CHAPTER 4

PRESENTATION AND INTERPRETATION OF RESULTS

4.1 Model Accuracy Comparison



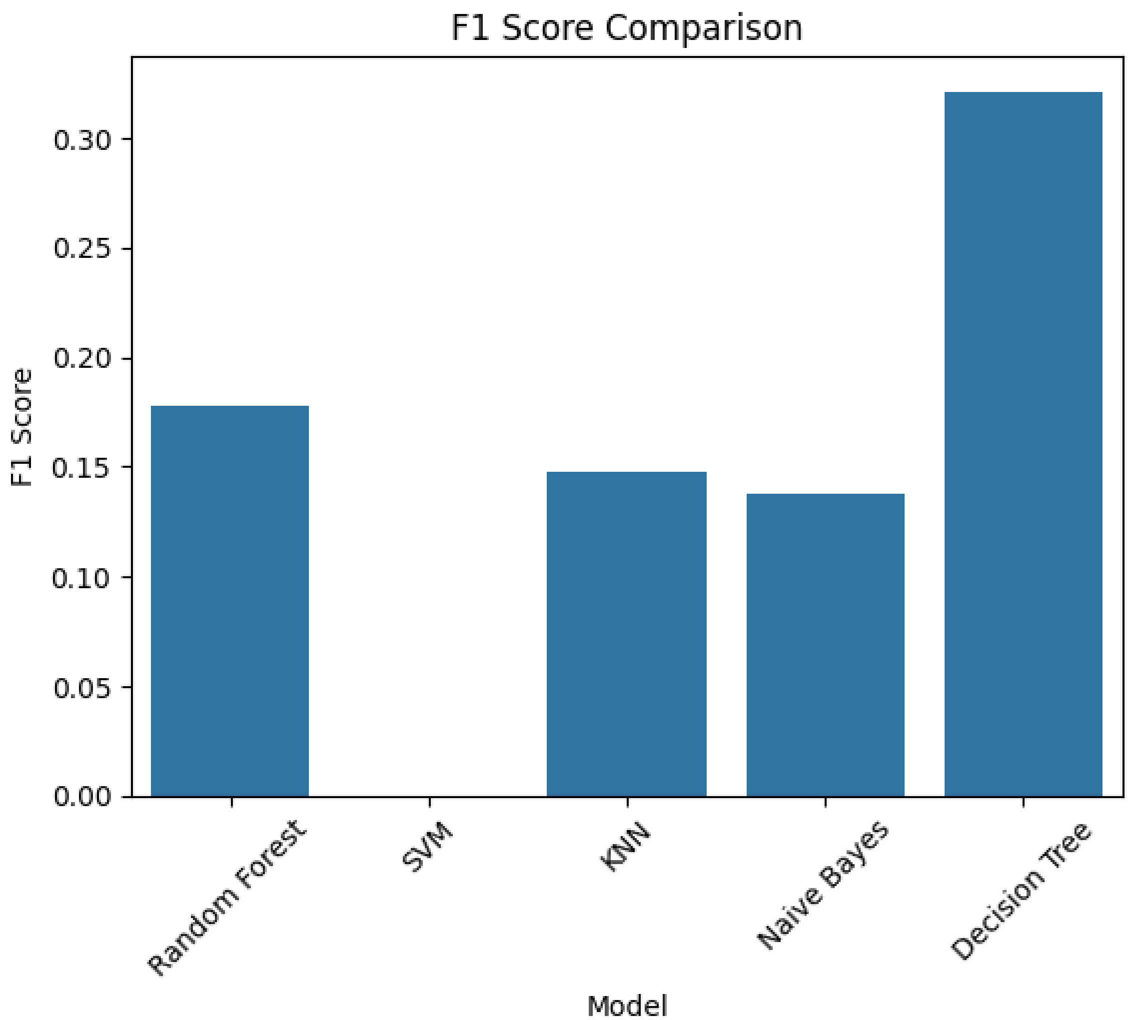
Interpretation

Shows how well each model predicts overall outcomes.

Inference

Higher accuracy ≠ best model (due to imbalance)

4.2 F1 Score Comparison



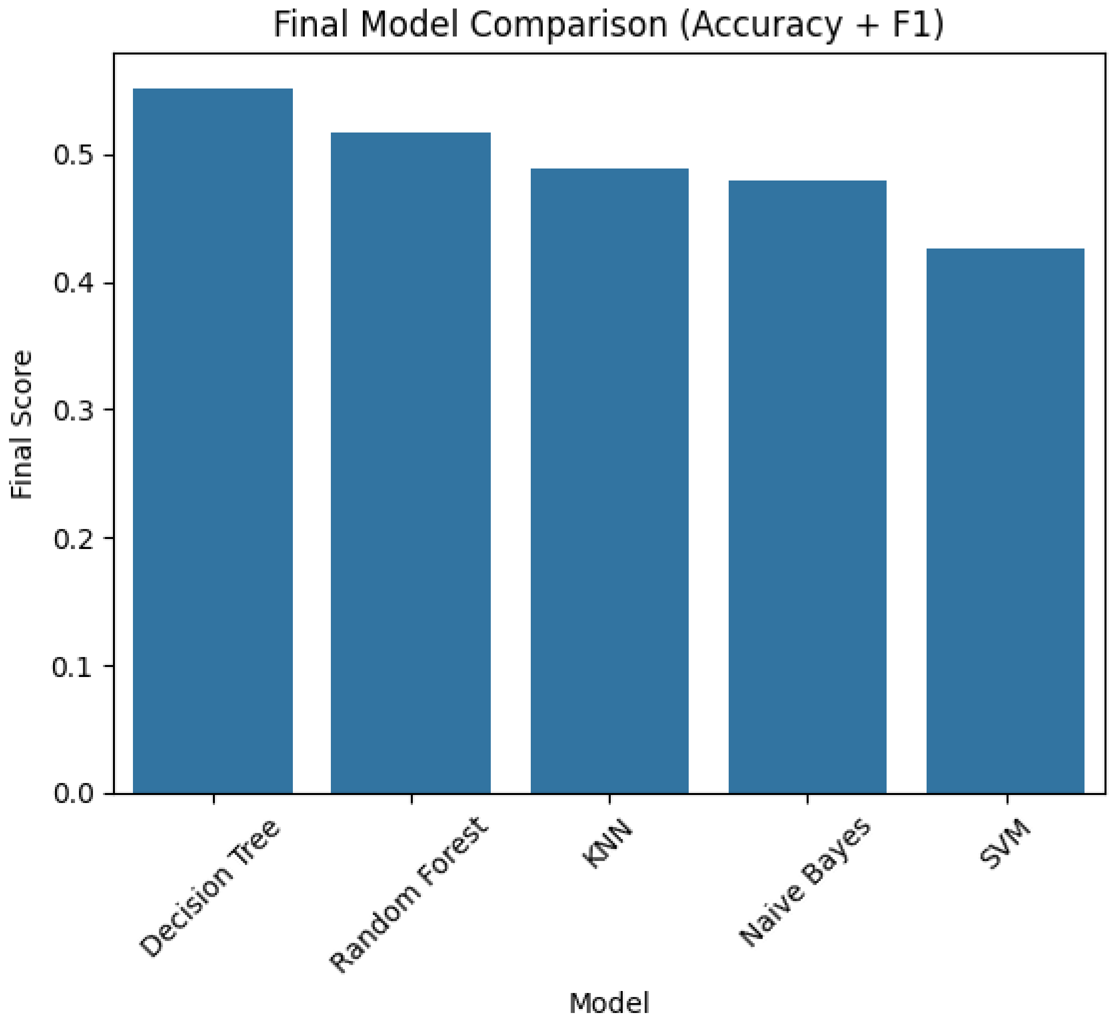
Interpretation

Measures balance between precision and recall.

Inference

Model with highest F1-score is more reliable in real-world scenarios.

4.3 Combined Performance Score



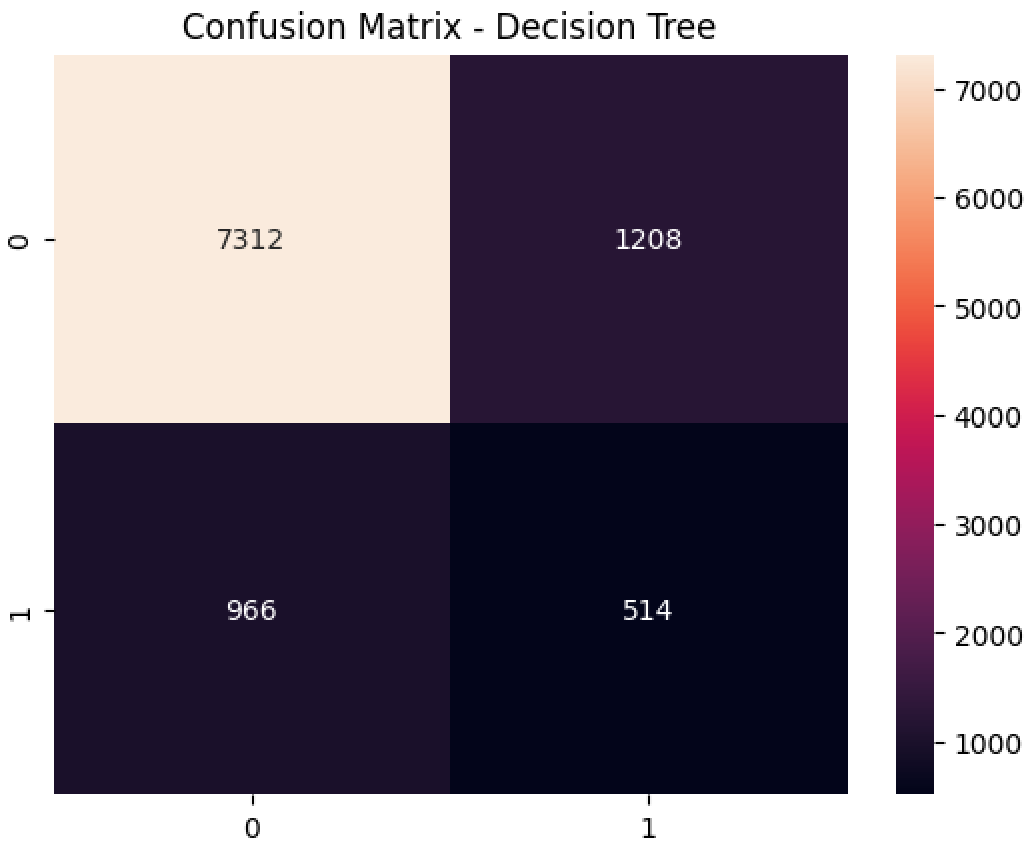
Interpretation

Combines accuracy and F1 Score for balanced evaluation.

Inference

The model with the highest score is most suitable for deployment.

4.4 Confusion Matrix



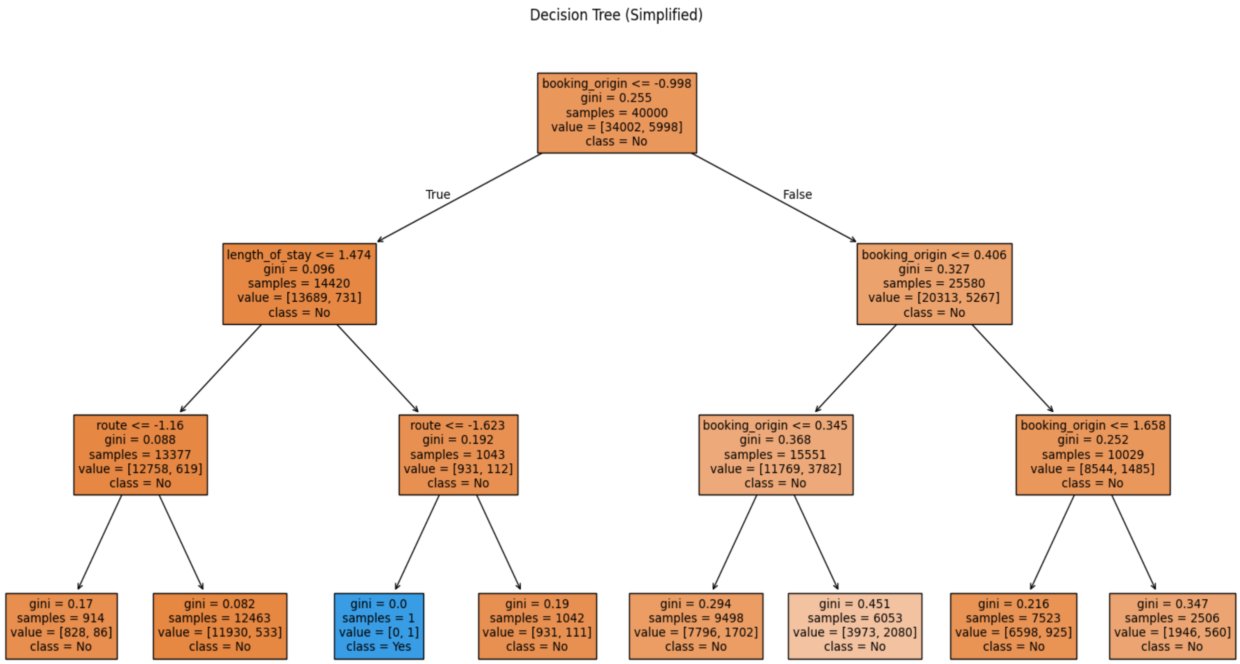
Interpretation

Shows correct and incorrect predictions.

Inference

Reducing false negatives is critical to avoid missing potential customers.

4.5 Decision Tree Visualization



Interpretation

Shows decision-making rules of the model.

Inference

Identifies key factors influencing booking completion.

4.6 Clustering Analysis

Clustering techniques applied:

- K-Means Clustering
- Hierarchical Clustering

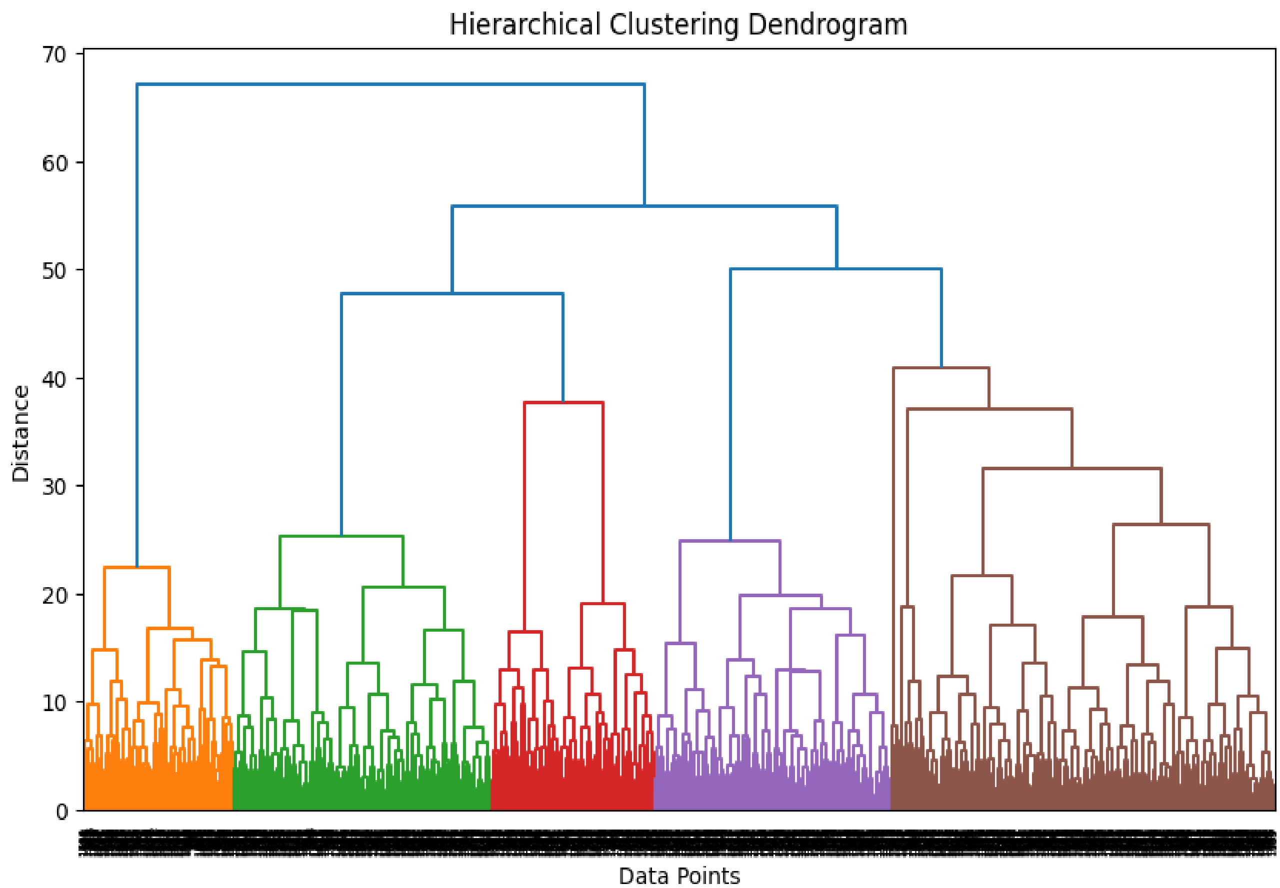
Interpretation

Customers are grouped based on behavioral similarities.

Inference

Helps identify distinct customer segments.

4.7 Dendrogram Analysis



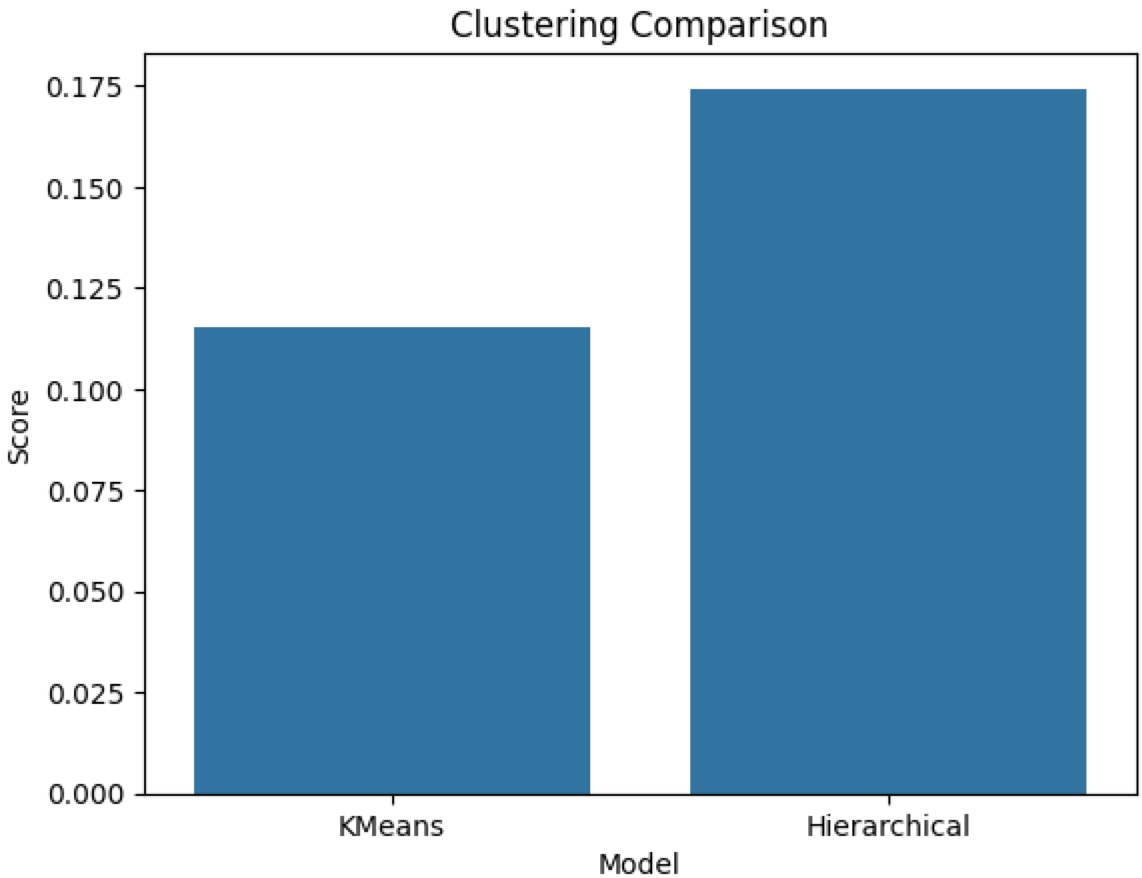
Interpretation

The dendrogram illustrates how data points are merged into clusters based on similarity. The vertical axis represents distance, and large gaps indicate natural cluster separations.

Inference

The optimal number of clusters is determined by identifying the largest vertical distance between merges. This indicates meaningful segmentation in the dataset.

4.8 Clustering Comparison



Interpretation

Compares clustering quality using silhouette score.

Inference

Higher silhouette score indicates better-defined clusters.

4.9 Best Clustering Selection

The clustering model with the highest silhouette score is selected.

Inference

Provides the most meaningful customer segmentation.

Hierarchical clustering and dendrogram analysis were performed on a representative sample to reduce computational complexity.

CHAPTER 5

MANAGERIAL / OPERATIONAL IMPLICATIONS

- Predict customer booking likelihood
- Improve targeted marketing strategies
- Enable customer segmentation
- Optimize resource allocation

These insights enhance:

  - Conversion rates
  - Customer satisfaction
  - Revenue growth

CHAPTER 6

CONCLUSION

This study applied classification and clustering techniques to analyze customer booking behavior.

Key Outcomes

- Identified best-performing classification model
- Evaluated models using multiple metrics
- Segmented customers effectively

Limitations

- Identified Class imbalance affects performance
- Sampling used for hierarchical clustering

Future Scope

- Use deep learning techniques.
- Apply advanced clustering
- Incorporate real-time data